Francisco Pinto Santos

Software Engineer | PhD



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About Me

Software Engineer with more than seven years of experience building and operating backend, distributed, identity and cloud-native platforms. Recent work spans consent management, CAMARA/Open Gateway integrations, observability and platform automation. PhD in Computer Science and Artificial Intelligence.

Skills

Backend and APIs: Java, Spring Boot, Node.js, Express.js, TypeScript, Python, Flask, Django, FastAPI and REST APIs.
Distributed data and architecture: Apache Kafka, Apache Spark, SQL, PostgreSQL, MySQL, MongoDB, Elasticsearch, Neo4j, microservices, distributed systems, data-driven architectures, Open Gateway and CAMARA.
Cloud platform and delivery: Bash, Git, GitHub Actions, CI/CD, Docker, Kubernetes, AKS, Helm, Helmfile, AWS, Azure and OpenFeature.
Observability: OpenTelemetry, Grafana, Loki, Mimir, Tempo and Prometheus.
Identity and ecosystem: OAuth 2.0, OpenID Connect, CIBA, JWT, JWS, JWE, JWK/JWKS, mTLS and Keycloak.
Frontend and documentation: React, Next.js, Docusaurus and MEVN/MERN.

Languages

Spanish: native.
English: full professional proficiency.

Experience

Software Engineer - Telefónica Innovación Digital

01/2024 - present · Madrid, Spain

- Owned Transparency Center, a React and TypeScript/Node.js component of a distributed consent-management platform. Implemented tenant-context consistency between decoded JWTs and request headers, plus backwards-compatible consent models for a multi-tenant migration.
- Owned a shared mock platform and led Node.js API, IdP and consent-management mocks for CAMARA APIs including SIM Swap and KYC, enabling integration and QA.
- Designed and validated REST API integrations and OAuth flows for partners, supported by test collections and technical documentation.
- Built internal administration and documentation portals with React, TypeScript, Next.js, Docker and Docusaurus.
- Owned branching strategies and automated validation, publication, synchronisation, build and deployment workflows.
- Built and evolved observability for a cloud-native AI computing platform. Implemented idempotent OTLP log replay, Kubernetes network policies, a development-environment end-to-end test harness and idempotent AKS cluster lifecycle automation.
- Owned identity tracks around API Gateway and contributed to AuthServer, implementing OAuth 2.0 and OpenID Connect (OIDC) consent flows and IdP integrations. Contributed to stronger authentication requirements and tenant-context propagation during the multi-tenant migration.

Support Engineer - Telefónica, BCNC Group

03/2023 - 01/2024 · Madrid, Spain

- Developed Node.js mocks and adapters for test-user and application provisioning, configurable data validation and OIDC behaviour. Tested and validated REST API and OAuth flows for third-party integrations within Open Gateway and CAMARA.
- Designed third-party integrations with the global consent-management platform, translating API, OAuth and identity requirements into mock behaviour, test collections and technical documentation. Diagnosed integration issues and validated features with platform and API teams.
- Operated integration environments and, with operations teams, investigated and resolved pre-production and production issues across backend microservices, Apache Spark and Apache Kafka. Contributed to consent flows and platform capabilities.
- Initiated and led support and environment-maintenance optimisation efforts, building internal flow-management tools, a Python CLI and CI/CD automation for data imports, cleanup and application lifecycle. Added retries to operational commands; informal estimates indicated roughly 50% less effort per support request and maintenance activity.

Backend & Architecture R&D Lead - BISITE Research Group

09/2020 - 03/2023 · Salamanca, Spain

- Designed and led the implementation of cloud-native R&D and AI platforms, including Apache Kafka-based ETL systems and internal tooling.
- Coordinated the technical delivery of an engineering team of around five people building platforms from scratch across ERP systems, scrapers, crawlers and AI-model APIs.

Full-stack Developer - BISITE Research Group

06/2019 - 09/2020 · Salamanca, Spain

- Owned and maintained the DeepInt Python SDK throughout its lifecycle, adding authentication, documentation, tests and higher-level workflows for users of the information-ingestion and AI-analysis SaaS.
- Built research and partner platforms from scratch, including Apache Kafka ETL systems and MEVN/MERN web applications. The research group operated DeepInt, which served more than **10K daily active users**.

Intern - BISITE Research Group

06/2017 - 06/2019 · Salamanca, Spain

- Developed REST APIs, MEVN web applications and Python data-curation tools, including crawlers and scrapers.

Lecturer - University and professional programmes

01/2021 - present · Salamanca, Spain

- **University of Salamanca:** taught Generative AI Project Design, Smart Cities Asset Management, IoT architectures and introductory artificial intelligence across postgraduate programmes.
- **Sociedad de Promoción Económica de Gran Canaria (SPEGC):** taught introductory artificial intelligence, convolutional neural networks and recurrent neural networks in 2022.
- **Pontifical University of Salamanca:** taught Database Management and Fundamentals of Computer Science I and II in 2022.

Education

PhD in Computer Science and Artificial Intelligence - University of Salamanca

10/2022 - 07/2026 · Salamanca, Spain

- Grade: *Summa cum laude*.

- Thesis: *Trustworthy Distributed Architectures for Intelligent Information Processing.*
- Research covered event-driven platforms with Kafka, Flink, NLP microservices, polyglot persistence, idempotent processing and privacy-by-design controls.
- Controlled thesis experiments sustained **100K events/minute** with **3 s p95 latency** for real-time NLP processing; a privacy-by-design evaluation over **5M events** achieved **99% audit coverage**.

MSc in Smart Cities & Intelligent Buildings - University of Salamanca

09/2021 - 09/2022 · Salamanca, Spain

- GPA: **9.04/10**.

MSc in Computer Engineering - University of Salamanca

09/2020 - 09/2022 · Salamanca, Spain

- GPA: **9.57/10**; first in the 2020-2022 class.
- Thesis: *DeepTrends: Distributed Platform for Information Retrieval and Processing*, a distributed real-time NLP architecture with event-driven processing and polyglot persistence.

BSc in Computer Engineering - University of Salamanca

09/2016 - 07/2020 · Salamanca, Spain

- GPA: **9.55/10**; first in the 2016-2020 class.

Achievements

- **04/2026:** third place in the University of Salamanca Alumni Talent Award.
- **01/2023:** best academic record award for the 2020-2022 MSc in Computer Engineering class at the University of Salamanca.
- **07/2021:** best academic record award for the 2016-2020 BSc in Computer Engineering class at the University of Salamanca.
- **07/2020:** third place in SEDEA's national ranking of engineering students.